



29th International Conference on Applications of Computer Algebra
ACA 2024

Program

University of Havana, Cuba
June 25 - 28

<https://math.unm.edu/~aca/ACA/2024/index.html>

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ACA 2024 - General Schedule

ACA 2024	Schedule			
Times	Tuesday 25	Wednesday 26	Thursday 27	Friday 28
9:00-9:30	Registration	COD	EDU	GEN
9:30-10:00		Plenary Talk Mijail Borges	EDU	GEN
10:00-10:30	Opening			
10:30-11:00			Coffee break	
11:00-11:30	Coffee break		INT	MI
11:30-12:00	IMA	COD		
12:00-12:30				
12:30-14:00	Lunch break			
14:00-14:30	EDU	IMA	INT	MI
14:30-15:00			PAN	
15:00-15:30	Coffee break	Excursion	Coffee break	
15:30-16:00	IMA		ACA WG Meeting	
16:00-16:30	Conference Photo			
16:30-17:00	Welcome Cocktail			
17:00-18:00				
18:00-19:00				
19:00-21:00			Conference Dinner	

EDU: *Computer Algebra in Education*

INT: *Integration and ODEs: Implementation and Applications*

COD: *Computer Algebra in Coding Theory and Cryptography*

IMA: *Computer Algebra for Digital Image Processing and Computer Vision*

MI: *Machine Intelligence and Symbolic Computation*

PAN: *Panel Discussion*

GEN: *General Session*

ACA 2024 - Plenary talk:

Wednesday, June 26th, 9:30-10:30

Mijail Borges Quintana

Study of algebraic and computational aspects of the Gröbner representations of linear codes.

Abstract: The research group of Computer Algebra of the Universidad de Oriente, Cuba, in collaboration with the University of Valladolid, Spain, has worked for several years on the study of Gröbner representations associated with linear codes and generalizations. The ideal associated with a linear code through the equivalence relation associated with the code, offers the possibility of defining and using the Gröbner bases of these ideals for problems related to the codes, such as decoding and determining the equivalence of codes. The Gröbner bases associated with linear codes have specific properties related to the codes. This structure and generalizations such as the Gröbner representations, the set of leader codewords and the complete Gröbner bases we show that can be computed from the formalization of a specialization of a Möller-type algorithm, using the computation in the finite-dimensional quotient algebra associated with the code. In addition, we studied some properties and structures related to the Gröbner bases of linear codes, such as the set of leader codewords of the code, its relationship with the test sets, correctable and uncorrectable errors of general linear codes, and the construction of invariants for linear codes. We show also how some of this structures and properties can be analyzed in the more general context of group codes, like lattice codes.

Special Session: Computer Algebra in Education.

Organized by Michel Beaudin, Michael Wester, Alkis Akritas, Noah Dana-Picard, José Luis Galán García and Elena Varbanova.

Tuesday, June 25th

Aula 2

14:00–14:30 Valentina Badía Albanés

The study of uniform convergence in the representation of functions using Computer Algebra Systems.

14:30–15:00 Włodzimierz Wojas

Can Chat GPT be a useful didactic tool in teaching and learning advanced mathematical analysis optimization topics?

15:00–15:30 Noah Dana-Picard and Zoltán Kovács

Estrella Solitaria in offset curves.

Thursday, June 27th

Aula 1

9:00–9:30 S. Bahrami and David J. Jeffrey

Ecstatic about extactic.

9:30–10:00 Michel Beaudin

Choosing an antiderivative is not easy!

10:00–10:30 J. Middeke, A. Olagunju and David J. Jeffrey

Database for Orthogonal matrices.

Special Session: Integration and ODEs: Implementation and Applications.

Organized by David Jeffrey, David Stoutemyer, Michael Wester and Michel Beaudin.

Thursday, June 27th

Aula 1

11:00–11:30 David J. Jeffrey and David Stoutemyer

Albert Rich 1949-2023

11:30–12:00 Michel Beaudin

Do like Rubi : don't be afraid of using $\operatorname{arctanh}$!.

12:00–12:30 A. C. Norman and David J. Jeffrey

A Future for Rubi?

14:00–14:30 Patrick Scheibe via Michel Beaudin

Rubi: Solving Integrals, One Rule at a Time.

Special Session: Computer Algebra in Coding Theory and Cryptography.

Organized by Ilias Kotsireas, Mijail Borges Quintana and Edgar Martínez Moro.

Wednesday, June 26th

Aula 1

9:00–9:30 **Daniel Cabarcas, Giulia Gaggero, Elisa Gorla**

The complexity of the SupportMinors Modeling for the MinRank Problem.

11:00–11:30 **Osmani Tito-Corrioso, Mijail Borges-Quintana, Miguel**

A. Borges-Trenard, Omar Rojas, Guillermo Sosa-Gómez

Combined General Methodologies of Key Space Partition for the Cryptanalysis of Block Ciphers.

11:30–12:00 **Ernesto Dominguez Fiallo, Pablo Freyre Arrozarena**

On the security of digital signature schemes based on CFS-like codes.

12:00–12:30 **Mijail Borges-Quintana, I. Alvarez-Barrientos, Miguel**

A. Borges-Trenard, E. Martínez Moro and J. A. Ornella Rodríguez

Complete Gröbner Basis for Linear and Lattice Codes.

Special Session: Computer Algebra for Digital Image Processing and Computer Vision.

Organized by Damian Valdés Santiago, Ángela M. León Mecías, Marta Lourdes Baguer Díaz-Romañach and José Alejandro Mesejo Chiong.

Tuesday, June 25th

Aula 1

11:30–12:00 Ángela M. León Mecías

Numerical and Image Analysis research group: lines of research, some results and work in progress.

12:00–12:30 Eric Martín-García and Damian Valdés Santiago

One-dimensional pattern detection using optimized and matched wavelets.

Tuesday, June 25th

Aula 1

14:00–14:30 Daniel García Pérez, Marta Baguer Díaz-Romañach and Manuel Vilas Valiente

Image inpainting using the smooth ordering of its patches: two new applications.

14:30–15:00 Nigreisy Montalvo Zulueta, Ángela M. León Mecías and Lorena León Arencibia

Texture analysis for breast mass malignancy classification in mammography images.

15:00–15:30 Lázaro Jesús Suárez Nuñez, Alba Álvarez González and Damian Valdés Santiago

Computational morphological analysis of digital images from irradiated tobacco seedlings.

16:00–16:30 José Alejandro Mesejo-Chiong and Leonardo Ulloa-Ferrer

3D reconstruction of Diabetic Foot Ulcers to evaluate healing progress.

Special Session: Machine Intelligence and Symbolic Computation.

Organized by Bruno Buchberger, Bo Huang, Dongchen Jiang, Dongming Wang and Jing Yang.

Thursday, June 27th

Aula 2

11:00–11:30 **Yang-Hui He**

The AI Mathematician: From Physics, to Geometry, to Number Theory.

11:30–12:00 **Rui-Juan Jing**

A Dataset for Suggesting Variable Orderings for Cylindrical Algebraic Decompositions.

12:00–12:30 **Dongchen Jiang**

Formal Representation of Mathematical Texts.

14:00–14:30 **Weifeng Shang**

A Framework of Plain Geometric Knowledge Discovery.

General Session

Organized by Michael Wester.

Friday, June 28th

Aula 1

9:00–9:30 **Dingkang Wang and Xiaopeng Zheng**

Completing Parametric Unimodular Rows to Unimodular Matrices.

9:30–10:00 **Bo Huang**

Algebraic Analysis of Zero-Hopf Bifurcations of Polynomial Differential Systems.

10:00–10:30 **Hoon Hong and Jing Yang**

Computing the greatest common divisor of several parametric univariate polynomials via generalized subresultant polynomials.

ACA 2024 - Panel Discussion:

Thursday, June 27th, 14:30-15:30

Aula 1

Organized by Jing Yang

****Overview****

The panel discussion at ACA 2024 aims to convene a distinguished group of researchers and practitioners to exchange knowledge and insights on pivotal topics within the realm of Computer Algebra. The scope of the panel will encompass a variety of subjects, including but not limited to:

- **Application Trends:** Engage in a dialogue about the most recent advancements, emerging trends, and groundbreaking innovations shaping the field of Computer Algebra and its practical applications.
- **Multidisciplinary Perspectives:** Elicit a spectrum of viewpoints by assembling a diverse panel of experts from varied disciplines, backgrounds, and specialized domains within Computer Algebra, ensuring a comprehensive discourse.
- **Future Directions:** Project and deliberate on the prospective trajectories and burgeoning areas that are poised to influence future research and development in Computer Algebra.

****Invitation to Participants****

We extend an open invitation to all ACA 2024 participants who are eager to contribute their perspectives on these topics and related areas. Join us for an engaging and enlightening discussion that promises to foster new connections and spark innovative ideas.